Advanced Financial Modeling, Module 4: 1-Week, Open-Ended Valuation and DCF Case Study (Jazz Pharmaceuticals)

In this module, you'll complete a 1-week, open-ended valuation and DCF case study based on Jazz Pharmaceuticals, a specialty pharmaceutical company that develops orphan drugs in the neuroscience, oncology, and epilepsy markets.

You'll start by projecting Jazz's revenue for its existing and pipeline drugs, including factors like the risk adjustment for drugs in Phase 2 clinical trials, royalties, and commercial milestone payments, and you'll learn how to research each drug's peak sales, market penetration, and pricing.

Then, you'll project the company's expenses and cash flow line items such as intangible purchases based on the assumed growth rates and the employee count.

Next, you'll learn how to complete Public Comps and Precedent Transactions "the real way" by manually reviewing annual and interim filings, adjusting for non-recurring charges, and calculating Equity Value and Enterprise Value in detail.

You'll conclude by building in more advanced DCF features, such as the mid-year convention, a stub period, a changing Discount Rate, a normalized Terminal Year, and a circular implied share price calculation, and you'll use these to build sensitivity tables and a valuation summary and graph.

AFM-04-01: Case Study Overview, Logistics, and Financial Model Setup (22:59)

In this introductory lesson, you'll learn what this Jazz Pharmaceuticals case study teaches, how to approach the revenue, expense, and cash flow projections, and which parts of the projections to focus on if you have a week to complete the case study.

Table of Contents:

- **3:58:** Part 1: The Biotech/Pharma Business Model
- **8:23:** Part 2: Recommended Approach and Initial Thoughts
- **14:56:** Part 3: Scenarios vs. Toggles
- **17:25:** Part 4: Components of Unlevered Free Cash Flow
- **21:10:** Recap and Summary

Quiz Questions:

14:56 Which of the following drugs would it make the most sense to project *in detail* in a valuation for a bio/pharma company?

a) Drug A (10% of revenue with ~2% average annualized growth).
b) Drug B (30% of revenue with the potential to grow by 20-30% over 5 years).
c) Drug C (15% of revenue with 8-10x growth potential).
d) Drug D (5% of revenue with 2x growth potential).

Explanation: The third answer choice (Drug C) is the best one here. Generally, you should make detailed projections for drugs with more significant revenue contributions and/or ones with high growth potential (or ones that are set to decline steeply). Therefore, you can eliminate Drugs A and D because they are fairly low percentages of revenue with less growth potential than Drug C. Drug B might also be a good candidate for detailed projections, but Drug C is better because 8-10x growth potential from a base of 15% of revenue indicates much higher potential future revenue than Drug B.

AFM-04-02: Xyrem Revenue Projections (16:02)

You'll learn how to forecast revenue for the company's key product, Xyrem, in this lesson, and you'll understand how to factor in uncertainty around the price increases and the entrance of generic drugs in this market.

Table of Contents:

- **3:10:** Data Gathering
- **6:29:** Market Size and Share
- **8:47:** Pricing
- **14:26:** Recap and Summary

Quiz Questions:

14:25 How might you determine the pricing growth for use in revenue projections for an established drug?

a) Look at historical price increases and extend the trend.
b) Look at price increases for comparable drugs in the market.
c) Back into the pricing increases based on "peak sales" figures and the projected patient count.
d) All of the above.

Explanation: Everything above is true. You can extend historical pricing trends, look at the price increases that comparable drugs have enacted, or back into price increases based on "peak sales" estimates from analysts and the projected patient population.

AFM-04-03: Xywav Revenue Projections (17:23)

In this lesson, you'll learn how to project revenue for the company's "replacement drug," Xywav, and how to determine the potential pricing and market penetration of an improved version of an existing drug.

Table of Contents:

- **0:59:** Data Gathering
- **7:49:** Pricing
- **9:41:** Market Size and Share
- **16:00:** Recap and Summary

Quiz Questions:

7:48 Which of the following factors might you assess to determine a company's pricing power with a new drug?

a) Its initial price vs. those of comparable drugs.
b) The level of branded and generic competition in the market.
c) How quickly the company has increased prices on other drugs historically.
d) All of the above.

Explanation: Everything above is true. You should consider all these factors when determining how a company might change its pricing and whether it is possible for the company to enact significant price increases for the drug.

AFM-04-04: Epidiolex Revenue Projections (15:42)

You'll learn how to gather historical revenue, pricing, and patient information for one of Jazz's acquired drugs, Epidiolex (a cannabinoid for seizure treatment), in this lesson, and you'll use these numbers to forecast the drug's market penetration, pricing, and sales.

Table of Contents:

- **0:34:** Data Gathering
- **6:50:** Marketing and Pricing Projections
- **14:14:** Recap and Summary

Quiz Questions:

6:50 Which of the following is the BEST reason to use peak sales estimates for a drug that are higher than estimates from analysts or industry news sites?

a) **The drug has been approved for additional uses after these estimates were published.**
b) You believe the analysts and news sites assumed market sizes that were too low.
c) You believe the company can raise its prices by 20% per year rather than the 5-10% that other sources assumed.
d) All of the above are equally justified.

Explanation: The first answer choice is the best one. Objectively, a drug that can be used to treat more diseases or conditions has a higher potential than one that can treat fewer diseases, so it's reasonable to assume higher peak sales if a drug is approved for additional indications. The other statements here are potentially plausible, but without further details, they're more difficult to justify than the first statement. For example, if the company's pricing power is 2-4x higher than what analysts have assumed, what market data or pricing history supports this claim? Have insurers or government-run healthcare plans given any indication that they will approve prices in this range? Data like that would be needed to support this reasoning.

AFM-04-05: Rylaze, Sunosi, and Other Drugs – Revenue Projections (17:36)

In this lesson, you'll learn how to find the data for Jazz Pharmaceuticals' less-significant drugs and how to make projections for Rylaze (ALL), Sunosi (EDS), and existing/mature products such as Defitelio, Erwinaze, and Vyxeos.

Table of Contents:

- **2:22:** Rylaze Data and Projections
- **10:06:** Sunosi Data and Projections
- **14:12:** Other/Legacy Drug Assumptions
- **16:07:** Recap and Summary

Quiz Questions:

10:05 Which of the following represents a key challenge when you use the prices of other, similar drugs in the market to estimate the price of a newly approved drug?

a) Even if you can find the prices, you might not be able to find the proper dosage.
b) You may not be able to determine the length of treatment, so the annual price may be unclear.

c) It may not be clear whether insurers and governments will approve a similar drug at the same price for the same disease.
d) All of the above.

Explanation: All the statements above represent key challenges when using existing, similar drugs to estimate the price of a new drug. The biggest challenge is determining the average dose per patient and the average length of treatment; without those, information such as the price per vial or price per mg is not useful because you need the average annual price per patient in financial models.

AFM-04-06: Zepzelca Revenue Projections (Royalties and Milestone Payments) (17:52)

You'll learn how to project revenue for Zepzelca, a small cell lung cancer (SCLC) drug, in this lesson; the drug has a shrinking market as well as associated royalties and milestone payments owed to an outside partner (PharmaMar).

Table of Contents:

- **1:51:** Market Sizing and Pricing
- **6:21:** Peak Sales and Market Penetration
- **12:56:** Commercial Milestone Payments and Royalties
- **16:09:** Recap and Summary

Quiz Questions:

16:08 What is the MAIN difference in the accounting treatment of royalties vs. milestone payments for pharmaceutical companies?

a) Milestone payments are deducted on the Income Statement, while royalties are deducted on the Cash Flow Statement.
b) The royalty rate might change over time, but milestone payments are known quantities.
c) **Milestone payments are capitalized and amortized over time, while royalties are simple expenses on the Income Statement.**
d) The fair value of future royalty payments appears on the Balance Sheet, while the fair value of milestone payments does not.

Explanation: The third answer choice is correct. The company's Zepzelca investor presentation directly states the proper treatment for both these items. Royalties appear within COGS, while milestone payments are similar to CapEx, with a deduction on the CFS and amortization over their useful life. The first answer reverses this treatment, so it is incorrect. The second answer is wrong because the milestone payments are not "known quantities" – they might change over

time based on the drug's performance and the specific thresholds required to trigger payment. And the last answer is wrong because neither item appears directly on the Balance Sheet in "fair value" form. The capitalized milestone payments do, but there is no fair value adjustment for them, so they are shown at historical cost or book value.

AFM-04-07: Risk-Adjusted Revenue for Pipeline Drugs (JZP-385 and JZP-110) (20:25)

In this lesson, you'll project revenue from Jazz's pipeline drugs (JZP-385 and JZP-110) that are currently in Phase 2 clinical trials, and you'll learn how to make rough estimates for the market sizes, annual prices, and the risk-adjustment factor.

Table of Contents:

- **2:29:** JZP-385 Projections
- **9:50:** JZP-150 Projections
- **15:50:** Risk-Adjusting the Revenue
- **18:55:** Recap and Summary

Quiz Questions:

9:49 Why is it usually more useful to focus on projections for drugs in Phase 2 and Phase 3 clinical trials than Phase 1 and pre-clinical drugs in a biotech valuation?

a) Phase 2 and Phase 3 drugs tend to have higher potential revenue.
b) After applying a high risk-adjustment factor, revenue from even highly successful earlier-stage drugs will be quite small.
c) There is usually far more market and pricing information for Phase 2 and Phase 3 drugs.
d) All of the above.

Explanation: The second answer is the best choice. Since the risk-adjustment factor might be upwards of 80-90% for Phase 1 and pre-clinical drugs, the eventual revenue contribution in a model will be small, regardless of the drug's potential. The first answer is wrong because the potential revenue depends on the market size and the company's pricing power, not the drug's current testing status. And the third answer is wrong because there tends to be little information on the market and pricing for all pipeline drugs that are not yet approved, regardless of their current status in clinical trials.

AFM-04-08: Expense and Unlevered Free Cash Flow Projections, Part 1 (23:53)

You'll learn how to project COGS, Operating Expenses, Depreciation, Amortization, and Taxes for Jazz Pharmaceuticals in this lesson and complete the first part of its Unlevered Free Cash Flow calculations.

Table of Contents:

- **2:46:** COGS and OpEx
- **10:38:** Depreciation and Amortization
- **18:15:** Taxes, NOPAT, and Margins
- **22:01:** Recap and Summary

Quiz Questions:

18:12 Why is it useful to split up the Amortization of existing vs. new Intangible Assets in the Unlevered FCF projections for a biopharma company?

a) Because the Amortization of existing Intangibles will eventually reach $0.
b) Because the tax implications might potentially be different, depending on the source of the Intangible Assets.
c) Because you might need to apply error-checking to the amortization of New Intangibles when the useful life is less than the forecast period.
d) **All of the above.**

Explanation: Everything above is true. These all represent valid reasons to split up the Amortization based on the existing vs. new Intangible Assets. Amortization on "new" Intangibles will continue forever as long as the company keeps acquiring them – but we still have to make sure that more than the initial balance is not amortized – and the tax treatment may differ for internally generated vs. acquired Intangibles. This last point doesn't necessarily apply to all companies, as Intangibles usually come from acquisitions – but existing Intangibles are more likely to be internally generated.

AFM-04-09: Expense and Unlevered Free Cash Flow Projections, Part 2 (16:10)

In this lesson, you'll finish up the UFCF projections for Jazz, estimate its Normalized Net Income to Common, and make a few other adjustments to the Xyrem projections to get more sensible numbers based on our outside research.

Table of Contents:

- **2:05:** Cash Flow Line Items
- **9:28:** Normalized Net Income to Common

- **12:33:** Other Adjustments
- **14:47:** Recap and Summary

Quiz Questions:

9:27 Is it acceptable for a company's Unlevered FCF growth rate to be negative in the final year of the DCF projections?

a) No – the company's cash flow should always be growing into the Terminal Period.
b) Yes – it's common to make this assumption for mature companies.
c) It depends on the company and its market and business model.

Explanation: The last answer is the best choice. This one depends completely on the company and its business model; if it's a company with "expiring" resources such as patents or oil fields that are no longer economically viable after a certain point, it's reasonable to assume declining UFCF in the projections. On the other hand, if the company could grow forever as it adds new locations or increases its customer count, and it doesn't have to deal with the same "expiring resource" issue, then its UFCF growth might be slightly positive by the end of the forecast period. There's no universal answer, which is why the DCF assumptions here differ significantly from the ones for Steel Dynamics in the previous module.

AFM-04-10: Screening for Public Comps and Precedent Transactions (17:32)

You'll learn how to screen for comparable public companies and precedent transactions for Jazz in this lesson, including what to do if you must rely on free sources because you don't have access to paid services such as Capital IQ.

Table of Contents:

- **1:26:** Public Comps
- **7:32:** How to Find Information via Free Sources
- **10:37:** Precedent Transactions
- **15:46:** Recap and Summary

Quiz Questions:

10:36 Which of the following is a DOWNSIDE of using free online data sources for the public comps rather than finding the data yourself or using a paid service?

a) It may be more difficult to screen the companies by useful criteria.

b) Free sources may not include the full set of companies you could find in paid databases.
c) The numbers may be less accurate because free sources may not adjust for non-recurring charges.
d) **All of the above.**

Explanation: Everything above is true. These statements all represent the downsides of using free sources: screening can be more difficult, the data and company sets may be incomplete, and the numbers may be less accurate.

AFM-04-11: Gathering Market Data for the Public Comps – BioMarin Example [OPTIONAL] (29:01)

In this lesson, you'll complete the first part of "spreading the comps" for BioMarin Pharmaceutical [BMRN], and you'll learn how to find information on the share count, dilutive securities, and line items in the Enterprise Value bridge.

Table of Contents:

- **2:00:** Why "Spread the Comps"?
- **5:00:** Model Setup and Process
- **9:51:** Share Price and Dilution Information
- **19:56:** Enterprise Value Bridge
- **27:24:** Recap and Summary

19:56 If a comparable company has convertible bonds outstanding, how do you count them in an analysis of the public comps?

a) Compare the conversion price to the current stock price and treat them as debt or equity based on that.
b) If the bonds count as equity, add back the cash interest expense and discount and fee amortization in metrics like Net Income.
c) If the bonds count as debt, use their fair market value in the Enterprise Value bridge, if possible.
d) All of the above.

Explanation: Everything above is true. Convertible bonds may count as either debt or equity depending on their conversion price relative to the company's share price. If they count as equity, you should add back the cash interest expense and amortization line items associated with them that reduce the company's Pre-Tax Income and Net Income when calculating Net

Income. If they count as debt, it's best to use their fair market value as debt instruments in the bridge.

AFM-04-12: Finding Financial Figures for the Public Comps – BioMarin Example [OPTIONAL] (32:29)

You'll calculate the LTM Revenue, EBITDA, and Normalized Net Income to Common for BioMarin in this lesson, and you'll learn how to adjust for non-recurring charges and the company's convertible bonds (as the cash interest and debt discount and fee amortization will change if the convertibles count as traditional debt).

Table of Contents:

- **1:49:** Part 1: LTM Revenue & EBITDA
- **11:25:** Quarterly Figures
- **13:17:** Part 2: Normalized LTM Net Income
- **22:19:** Convertible Bond Adjustments
- **26:26:** Taxes and Net Income to Common
- **28:56:** Part 3: Projected Figures
- **29:53:** Part 4: More Practice
- **31:05:** Recap and Summary

Quiz Questions:

13:16 Which of the following items would you be MOST likely to add back as a non-recurring charge when calculating EBITDA?

a) A Restructuring expense that recurs in 2 / 3 historical years.
b) An Inventory Reserve that significantly increases COGS in 1 / 3 historical years.
c) An Impairment Charge that reduces Pre-Tax Income in 1 / 3 historical years.
d) A Change in the Fair Value of Contingent Consideration that reduces Operating Income in 3 / 3 historical years.

Explanation: The second answer choice is the best one. You can rule out the first and last answer choices because if an expense occurs in 3 / 3 or 2 / 3 historical years, it's unlikely to be "non-recurring." The threshold is usually closer to 1 / 5 years, which immediately disqualifies these choices. The third choice is incorrect because this Impairment Charge affects only Pre-Tax Income – but the EBITDA calculation starts with Operating Income, which is above Pre-Tax Income. Therefore, by process of elimination, the second answer is the best choice.

AFM-04-13: Calendarization Overview and Public Comp Analysis of Jazz Pharmaceuticals [OPTIONAL] (21:16)

In this lesson, you'll learn the key points required to calendarize companies' financials, and you'll learn where the information required to calculate Equity Value, Enterprise Value, and the valuation multiples for Jazz Pharmaceuticals comes from.

Table of Contents:

- **0:56:** Calendarization
- **6:05:** Jazz – Diluted Shares
- **9:00:** Jazz – TEV Bridge
- **11:14:** Jazz – LTM and Projected Financials
- **19:17:** Recap and Summary

Quiz Questions:

19:16 If there's a non-recurring expense that is NOT shown explicitly on the company's financial statements, what's the most efficient way to determine where it might be?

a) Read through the full notes to the financial statements.
b) Search for "non-recurring expense."
c) Check to see if any of the normal expense categories are unusually high in one period.
d) Look at equity research and check the adjustments that analysts made in the historical periods.

Explanation: The third answer choice is the best one here. Few companies disclose actual non-recurring expenses as "non-recurring expenses" in their filings, so the second choice is unrealistic. And while you can find these charges by reading through the full notes to the statements, it's an extremely inefficient method. The last answer is more reasonable, but it assumes that you have access to equity research and that the research analysts have adjusted the historical numbers using standards that match your own, which is not always true. So, the third answer is the most efficient way to tell at a glance if the company might have a non-recurring expense embedded in one of its line items.

AFM-04-14: Precedent Transaction Analysis of AstraZeneca / Alexion Pharmaceuticals [OPTIONAL] (20:42)

You'll learn how to gather data and fill in the Equity Value, Enterprise Value, and LTM and projected financial figures for one of the precedent transactions in this lesson: AstraZeneca's acquisition of Alexion Pharmaceuticals.

Table of Contents:

- **2:34:** Offer Price and Share Prices
- **4:27:** Diluted Shares
- **7:28:** TEV Bridge
- **12:43:** LTM Revenue and EBITDA
- **17:14:** Projected Financials
- **19:28:** Recap and Summary

Quiz Questions:

12:43 Which of the following represents a DIFFERENCE in how you calculate Equity Value and Enterprise Value for a precedent transaction (vs. a public comp)?

a) For the transaction, you might get the share count and diluted share information from the deal documents, such as the merger agreement.
b) You use the offer price per share rather than the company's current share price for the transaction.
c) You might use options outstanding rather than options exercisable because unvested options typically vest when there's a change of control.
d) All of the above.

Explanation: All these answer choices represent DIFFERENCES in the "detailed calculations" for a precedent transaction vs. a public comp. Documents beyond the 10-K and 10-Q, such as the Fairness Opinion and merger agreement, are more important, and all the multiples should be based on the offer price rather than the company's current share price. And it is technically better to use options outstanding rather than options exercisable in the diluted share calculation for the reason stated here.

AFM-04-15: Cost of Debt Calculation with Convertible Bonds (18:12)

In this lesson, you'll learn how to calculate the Cost of Debt for companies with convertible bonds and companies with a mix of convertible bonds and traditional debt, such as term loans and senior secured notes.

Table of Contents:

- **3:32:** BioMarin Calculation
- **10:10:** Jazz Calculation
- **16:36:** Recap and Summary

Quiz Questions:

10:10 When a company has convertible bonds, which of the following is NOT required to estimate its Cost of Debt?

a) The maturity date of the bonds.
b) The current market value or "fair value" of the bonds.
c) The face value or principal amount of the bonds.
d) The cash coupon rate on the bonds.
e) The effective interest rate on the bonds.

Explanation: The fourth answer choice is correct, i.e., it is NOT required to estimate the Cost of Debt for a company with convertible bonds. You do not use this cash coupon rate in the YIELD function at all because if the convertible bonds count as traditional debt, you should use the coupon rate on equivalent non-convertible debt, otherwise known as the "effective interest rate," so the actual cash coupon rate is unnecessary. Everything else is a required input into the YIELD function.

AFM-04-16: WACC Calculations (20:28)

You'll complete the WACC calculations for Jazz in this lesson and use several different methods, as well as the WACC figures for the comparable companies, to determine the proper range of values for use in the DCF.

Table of Contents:

- **2:16:** Step 1: Pulling in the Data
- **7:26:** Step 2: WACC for the Comparables
- **9:39:** Step 3: Un-levering and Re-levering Beta
- **14:26:** Step 4: Cost of Equity and WACC
- **18:51:** Recap and Summary

Quiz Questions:

18:50 If there is a big difference between the Debt and Equity percentages in a company's current capital structure vs. those in the median capital structure of the comparables, why might that not affect the WACC range by much?

a) The lower Equity % in one case may be offset by the higher Cost of Equity (due to added risk from leverage).

b) The Costs of Equity and Debt and the percentages of Equity and Debt are not necessarily related.
c) Companies with less Equity tend to use more Preferred Stock, and its higher cost offsets the reduced Equity contribution to WACC.
d) All of the above.

Explanation: The first answer choice is the best one. All else being equal, if the same company goes from a high Equity percentage (90%+) to a much lower one, such as 50-60%, its Cost of Equity and Cost of Debt will both increase, but its Cost of Equity will tend to increase by a greater percentage because of the significant added risk from leverage. So, in the WACC formula, which is based on the percentage of each form of capital times its cost, the reduced Equity percentage may be offset by the higher Cost of Equity. The second answer is false because the percentages of Equity and Debt very much affect their costs, as added leverage increases the risk for all investor groups, and the third answer makes no sense because Preferred Stock is the "Plan C" option for most companies and is uncommon if the company can still issue traditional Debt.

AFM-04-17: The Mid-Year Convention and Stub Periods (17:06)

In this lesson, you'll learn about the factors that can make it more complicated to discount the cash flows in a valuation, such as stub periods, the mid-year convention, and a changing Discount Rate, and you'll set up the normal and mid-year discount periods to support these features.

Table of Contents:

- **2:40:** Set Up the Assumptions
- **4:29:** Set Up the Discount Periods
- **7:42:** Set Up the Mid-Year Convention
- **11:09:** Practice Exercises
- **15:20:** Recap and Summary

Quiz Questions:

11:08 In Year 1 of a DCF analysis, the stub period, before the mid-year convention, is 0.600. What is the discount period for Year 3, factoring in both this stub period and the mid-year convention?

a) 1.300.
b) 2.300.

c) **2.100.**
d) 1.800.

Explanation: The third answer choice is correct. To calculate the discount period, you take the entire stub period in the first year (0.600), add the discount period for the entire second year (1.000), and then add half the discount period for the entire third year (0.500) to reflect the mid-year convention. 0.600 + 1.000 + 0.500 = 2.100.

AFM-04-18: A Changing Discount Rate and Cumulative Discount Factors for Unlevered Free Cash Flow (14:57)

You'll calculate the cumulative Discount Factors under both the normal and mid-year conventions in this lesson, and you'll use them to discount the company's UFCF to Present Value in each year, factoring in the portion of Year 1 that has already passed.

Table of Contents:

- **2:16:** Part 1: The Changing Discount Rate
- **3:44:** Part 2: The "Remainder" Cash Flow
- **6:57:** Part 3: The Cumulative Discount Factors
- **11:08:** Part 4: PV of Unlevered FCF
- **12:56:** Recap and Summary

Quiz Questions:

12:54 Which of the following features makes it necessary to use the Cumulative Discount Factor to discount cash flows in a DCF rather than the built-in NPV function?

a) A Changing Discount Rate in the forecast period.
b) A stub period in Projected Year 1.
c) The mid-year convention.
d) All of the above.

Explanation: Everything above is true. The NPV function only works when the Discount Rate is constant, and each cash flow is separated by an even 1-year interval. The stub period and mid-year convention mean that some of the intervals are no longer one year apart, and the changing Discount Rate over this forecast period cannot be accepted as input into the NPV function.

AFM-04-19: The Normalized Terminal Year (15:32)

In this lesson, you'll calculate the Normalized Terminal Year for Jazz, which represents Year 1 of the Terminal Period, and you'll learn why it can be useful when the company's financial characteristics change significantly from the end of the forecast period into the Terminal Period.

Table of Contents:

- **2:15:** Part 1: Revenue Growth Rate and Margins
- **7:51:** Part 2: NOPAT and Cash Flow Line Items
- **11:42:** Part 3: UFCF and EBITDA
- **13:00:** Part 4: Normalized Terminal Year Impact
- **14:14:** Recap and Summary

Quiz Questions:

14:14 What is the MOST LIKELY adjustment when you calculate the Normalized Terminal Year in a DCF?

a) You usually assume a higher revenue growth rate.
b) You often assume lower operating margins.
c) You often assume that items that eventually "expire" disappear completely.
d) All of the above.

Explanation: The third answer choice is the best one. Depending on the company and industry, the growth rate and margin could be higher or lower in the Normalized Terminal Year; there isn't necessarily one "correct" answer or most common adjustment. However, you almost always eliminate non-recurring items and items that eventually expire, such as the Amortization of Intangibles, in the Normalized Terminal Year.

AFM-04-20: Terminal Value Calculations Including the Mid-Year Convention and Stub Periods (13:15)

You'll calculate the Terminal Value and the Present Value of the Terminal Value in this lesson, factoring in the mid-year convention, stub period, normalized Terminal Year, and the changing Discount Rate and cumulative discount factors.

Table of Contents:

- **2:07:** Step 1: Terminal Growth Rates and Terminal Multiples
- **6:54:** Step 2: Implied Terminal FCF Growth Rate
- **8:21:** Step 3: PV of Terminal Value

- **11:59:** Recap and Summary

Quiz Questions:

11:58 Which of the following points CHANGES in the PV of Terminal Value calculation when you use the mid-year convention and a stub period in the DCF?

a) **The discount period under the Perpetuity Growth Method reflects both of these, but the Multiples Method only reflects the stub period.**
b) You must use a different discount period for the PV under the Perpetuity Growth Method, but not the Multiples Method.
c) It depends on whether or not the Discount Rate changes over the forecast period.
d) None of the above.

Explanation: The first answer is correct. The Perpetuity Growth Method is based on cash flows, and in a DCF with the mid-year convention, all cash flows are assumed to arrive midway through each year. Therefore, the Terminal Value is also midway through the final forecast year, so we use a discount period that is half a year earlier (e.g., 9.5 rather than 10.0). If there is also a stub period in the analysis, that factors into the PV of Terminal Value under both methods because the partial year in the beginning is there regardless of how you calculate Terminal Value.

AFM-04-21: Implied Share Price Calculation and Sensitivity Tables (18:10)

In this lesson, you'll set up an implied share price calculation that includes circular references due to the company's dilutive securities, and you'll create sensitivity tables to analyze the valuation output under different assumptions.

Table of Contents:

- **1:32:** Implied Share Prices
- **10:21:** Sensitivity Tables
- **16:50:** Recap and Summary

Quiz Questions:

10:20 Which of the following forms of dilutive securities makes it MOST important to make the Implied Share Price calculation in a DCF circular?

a) **Convertible bonds.**

b) Stock options and warrants.
c) Restricted stock units (RSUs).

Explanation: The first answer is the best choice. The last answer is wrong because RSUs are a straight addition to the diluted share count and are not affected by the implied share price in this calculation. The dilution from stock options and warrants is affected by the company's implied share price, but it's a very small effect compared with convertible bonds because convertible bonds are counted as "all or nothing" – either the entire issuance converts into shares, or none of it does. Also, convertible bonds affect the Enterprise Value bridge because they do not count as debt if they convert into shares.

AFM-04-22: Valuation Summary ("Football Field" Graph), Model Summary, and Cover Sheet (15:48)

In this final lesson of the module, you'll learn how to create the valuation summary sheet, factoring in circular references for the implied share price calculations, and you'll interpret the valuation and see why a model summary page and cover sheet can be useful.

Table of Contents:

- **1:19:** Valuation Summary
- **9:24:** Valuation Graph
- **11:19:** Model Summary and Cover Sheet
- **14:01:** Recap and Summary

Quiz Questions:

[No quiz questions.]